



Canopy Disturbance Check · Browser

Sentinel-2 NDVI, NDMI and NBR change detection for ACR IFM validation and verification, run from a web page. **No install, no account, no Cloud project, no billing.**

SOP CHEAT SHEET

Prototype Tools
Checker 0.13.1 · Viewer 0.2.1
SOP of 3 September 2026

1 Requirements

NO INSTALL, NO ACCOUNT

No Python environment, no plugin manager, no elevated shell, no ASRA. Any current browser serves, **Edge on a managed Windows 11 machine at standard user permissions included**. Steps 1 and 2 of the QGIS and ArcGIS SOPs do not exist here. There is no sign-in, no OAuth client, no Cloud project bound for metering, no billing and no one-hour session.

OUTBOUND HTTPS

A restricted workstation needs `prototype-tools.github.io` for the application, `earth-search.aws.element84.com` and `sentinel-cogs.s3.us-west-2.amazonaws.com` for the scene catalogue and the pixels, `services3.arcgis.com` for fire perimeters and the aerial detection survey, `apps.fs.usda.gov` for the activity tracking record, `lfps.usgs.gov` and `imagery.geoplatform.gov` for LANDFIRE and LCMS, `power.larc.nasa.gov` for the climate series, `cwfis.cfs.nrcan.gc.ca` for Canadian burned area, and `s3-us-west-2.amazonaws.com` with `services.arcgisonline.com` for the dated imagery archive.

NO DATA-HANDLING EXEMPTION IS REQUIRED Uploaded site data is parsed in the browser and never leaves the workstation.

2 Imagery

PROVENANCE, NOT A CONTROL

Scenes are found through Element 84's public **Earth Search** catalogue and read out of the Copernicus **Sentinel-2 Level-2A surface reflectance** cloud-optimised GeoTIFFs on AWS Open Data, by **HTTP range request at 20 m**, so only the pixels the analysis needs are fetched.

THE PRE-2022 BASELINE PROBLEM CANNOT ARISE The +1000 DN bottom-of-atmosphere offset is removed **PER SCENE**, from the catalogue's `earthsearch:boa_offset_applied` flag rather than inferred from the acquisition date, because reprocessed old scenes carry it. Scenes either side of the January 2022 processing baseline are radiometrically comparable.

TWO CLOUD MASKS

OmniCloudMask, a segmentation model run in the tab, decides from shape and texture, which is how a person separates cloud from bright ground and cast shadow from a north-facing slope. **The Sen2Cor scene classification layer** ships with every Level-2A product and supplies snow, saturated and no-data either way, but alone it misses thin cloud edge and labels topographic shade as cast shadow.

On a **61 per cent cloudy** Blackfeet overpass of 27 August 2024, OmniCloudMask called **11.4 per cent** of the block cloud or shadow where the scene classification called it clear, against **0.75 per cent** the other way. Thin cloud edge and shadow leaking into a composite read as canopy loss, so the asymmetry is the point.

Either mask feeds a **per-pixel median** over surviving observations. Water is masked at the delta stage, and a pixel that is water in **either** window is masked in both, so a lake against a mudflat reads as a water-level change.

3 Analysis extent

THREE ROUTES, IN ORDER

- **Upload the project boundary.** The most reliable. Loaded under Site data it becomes both the analysis extent and a drawn outline.
- **Bounds.** West, south, east, north in EPSG:4326, or **Use current map view.** Reversed pairs are normalised, so a swapped east and west cannot silently return an empty geometry, which the `raw ee.Geometry.Rectangle` call in the scripts accepts without error.
- **GeoJSON.** Paste a geometry.

Which tool when

THREE ROUTES

Browser for a screening pass with no install, and for the season charts the script SOPs send you here for. **QGIS** for Geemap with layers drawn by Earth Engine and GeoTIFF export to Drive. **ArcGIS Pro** for an archival GeoTIFF, or work that must stay in an existing Pro project.

Buffer before upload

NO BUFFER_M HERE

A margin of roughly **500 m** earns its place, because a hard clip at the legal boundary cuts off the far half of a cut block straddling the line and hides leakage onto adjacent ownership, the subject of Appendix A.4.

THE CHECKER HAS NO BUFFER CONTROL The QGIS and ArcGIS scripts do this for themselves through `BUFFER_M`. Buffer in QGIS or Pro on a **BEVEL** join, save as a separate file, upload that one. Simplify at 10 m, the pixel size, first, because buffering offsets every vertex outward and carries every one through.

Name it so it cannot be mistaken for the boundary, `Boundary_buffer500m.shp`. **Every hectare figure is read off the unbuffered boundary**, the reporting geometry the scripts hold separately as `roi_report`. Buffering a 4,712 ha test parcel by 500 m returned **6,067 ha**, an inflation of 29 per cent, and an area quoted from it cannot be defended against a registry acreage.

Run on the buffered file and load the unbuffered boundary as an outline under Site data, so a reader sees which ring a number came from. **Disturbance between the two rings is reported as adjacent, never as project.**

4 Reporting periods

FOUR DATES EACH

The start and end of the **pre** window and of the **post** window, per Reporting Period. Each window is a range from which one cloud-free composite is built, not a single acquisition. Defaults are **1 August to 1 September**, matching the production scripts; a July to September growing-season window is acceptable. Set the years to the Reporting Period under verification, pre at its start, post at its end. **Add reporting period** chains a further period, and all periods share one set of class breaks, so differences between periods are real rather than an artefact of settings.

DORMANT-SEASON COMPOSITES CORRUPT DNDMI Senescence draws SWIR1 reflectance up before leaf-fall and gives a uniform false moisture-stress signal. In Appendix A.1 of the findings SOP a reporting calendar drifted out of the growing season and flagged **38 PER CENT** of a project area, which July to September windows collapsed to **4 PER CENT**, all of it on beetle survey polygons.

Confirm the season

THREE CHECKS, THEN RUN

Once an extent is set the panel draws the daily climate at its centre from **NASA POWER**, air temperature, sunlight, rain over the previous four weeks and, where any fell, snow depth. Each period year is a coloured line over a grey ten-year mean, windows shaded, and the table beneath gives the per-window means that go to the manifest.

Same dates one shaded band	Same season lines on the mean	Same ground no lone snow
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Same dates. Two bands mean the windows sit on different parts of the year. A March pre window against an October post window differed by **11.5 °C** and **270 mm** of rain, and the delta would have reported a Mediterranean summer drying as disturbance.

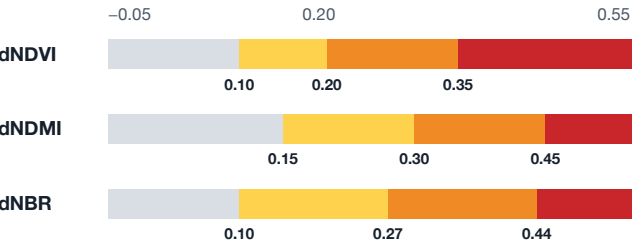
Same season. A line below the mean in spring is a late year, above it an early year, and a deciduous canopy at 1 April in a late year can run three weeks behind. Prefer the August default, where the summer plateau makes a fortnight matter least.

Same ground. No snow in either window, rain within the ordinary range of the years drawn. Where that cannot be met, expect a uniform shift in the dNDMI histogram, read it as season rather than stress, and say so in the finding.

The grid is about fifty kilometres across, so it is the season of the district. Where a year ran hot and dry, as August 2022 did over Montana at **3.2 °C** warmer and half as wet as August 2021, **the window stays where it is and the finding says why the period differs.**

5 Class breaks

GOVERNING DEFAULTS



Class breaks on a common scale. Grey undisturbed, yellow Low, orange Moderate, red High. These are the governing defaults.

0 Undisturbed masked out	1 Low screen	2 Moderate inspect	3 High draft finding
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dNDVI and dNDMI from SOP Step 6, dNBR from MTBS and USFS PNW; **where the SOPs disagree the browser tools follow Step 6**. Ordering is enforced, and any value moved off its default marks that index adjusted and requires a written justification in the manifest.

Applying a moved break re-runs the whole check, so move all three before applying.

6 Site data

NEVER UPLOADED

Project boundary, streamside management zones and plot points load from a zipped shapefile, GeoJSON or KML. Loading a boundary also sets it as the analysis extent. Plot points are labelled with their identifier, so a screenshot ties to an inventory record without a legend; the column is detected automatically, preferring `Plot ID` or `Plot_No` over `OBJECTID`. **Check the detected field.**

ZIP THE WHOLE SHAPEFILE The reader parses the `.prj` and reprojects to WGS84, so a UTM shapefile loads without preparation. A zip missing it has nothing to reproject from and its coordinates are read as degrees, which puts a Montana project in the Gulf of Guinea. **A BOUNDARY IN THE WRONG HEMISPHERE MEANS A MISSING .PRJ.**

7 Run

IN THE TAB

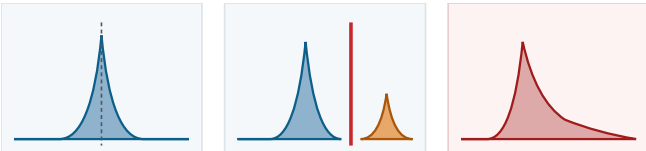
Press **Run check**. A small extent finishes under a minute; a large one, or several periods, takes longer, because pixels are fetched and the cloud model run in the tab rather than on a server. Success is **five layers per period**, three classified rasters visible on top, the pre and post true-colour composites hidden beneath, each carrying its period prefix.

8 Read results

HISTOGRAM BEFORE MAP

Panel section 6 gives overpass counts per window, three histograms with draggable break handles, and a class-area table in hectares on the **project UTM zone** at 20 m, EPSG:32612 on the Blackfeet run. The reprojection the QGIS SOP does by hand is automatic.

READ THE OVERPASS COUNTS FIRST Fewer than four clear looks per window makes the median unstable.



Defaults hold noise / phenology
Low break in gap real disturbance
Audit the dates cloud / season / BRDF

Read the shape before you touch a break. A long right tail with no gap is contamination, not disturbance.

Diagnostics cover dormant-season and mismatched windows, contaminated histograms, thin composites and thin coverage. **Each requires acknowledgement, recorded in the manifest.** Resolve every one before treating anything on screen as real, because a diagnostic usually means misleading imagery rather than a disturbed site. Template F.6 of the findings SOP carries the coverage caveat wording.

Quote the **share of the analysis extent** as well as the hectares, since it is more robust across scales, and areas here are at 20 m against 10 m in script exports.

Cross-check the indices

BEFORE DRAFTING

dNDVI and dNDMI are pre minus post, positive meaning loss; dNBR is post minus pre, positive meaning burned, per MTBS. Read the three together before drafting anything.

SIGNATURE	READ IT AS
dNDVI High + dNBR clean	Harvest, blowdown or clearing. Read the edge geometry.
dNDVI High + dNBR High	Fire. Confirm against the perimeter record in section 8 before raising.
dNDMI High alone	Moisture stress. Corroborate with the aerial detection survey before calling it disturbance.

Right angles and linear edges point to anthropogenic causes, harvest, road or right-of-way; curvilinear or amorphous edges to natural ones, blowdown, decline or slide. Both layers go into the screenshot package, because auditors and developers expect the cross-check rather than the single raster.

9 Visual check

WHAT THE ARITHMETIC CANNOT DO

Section 7 holds a crossfade between the pre and post true-colour composites over fixed ground, with a blink control. Both are built from the masked observations the analysis used, **not a single scene**, so what is on screen is what was measured. A clearing jumps between two dates; noise does not.

Beneath it, **Find available imagery** searches Esri's dated World Imagery archive for every distinct photograph of the site, replacing the Google Earth historical timeline the script SOPs lean on and needing no account.

QUOTE THE CAPTURE DATE, NEVER THE RELEASE DATE A release is when Esri published a version of the basemap; the photograph inside it was frequently taken a year or more earlier and varies from place to place inside one release. Citing a release date as a capture date is a factual error in a finding.

10 Corroboration

NEVER AN INPUT

Section 8 queries the independent record over the same extent and years the windows span. It returns mapped fire from the interagency perimeter feed, MTBS and the Canadian National Burned Area Composite; the Forest Service **aerial detection survey by year, agent, damage type and acreage**, which is what allows a dNDMI signal to be attributed rather than merely described; the management activity tracking record; LANDFIRE and LCMS change products as overlays.

AN EMPTY RETURN IS NOT PROOF The activity tracking record covers **NATIONAL FOREST SYSTEM LAND ONLY**, so on private, state or tribal trust ownership a blank return is an absence of jurisdiction, not of harvest.

Capture the survey table and the Layer panel showing which overlay is drawn and at what opacity, since a table alone does not show which years and product were queried. Nothing in the composite, delta, break or area depends on another party's model, which is what keeps the check independent.